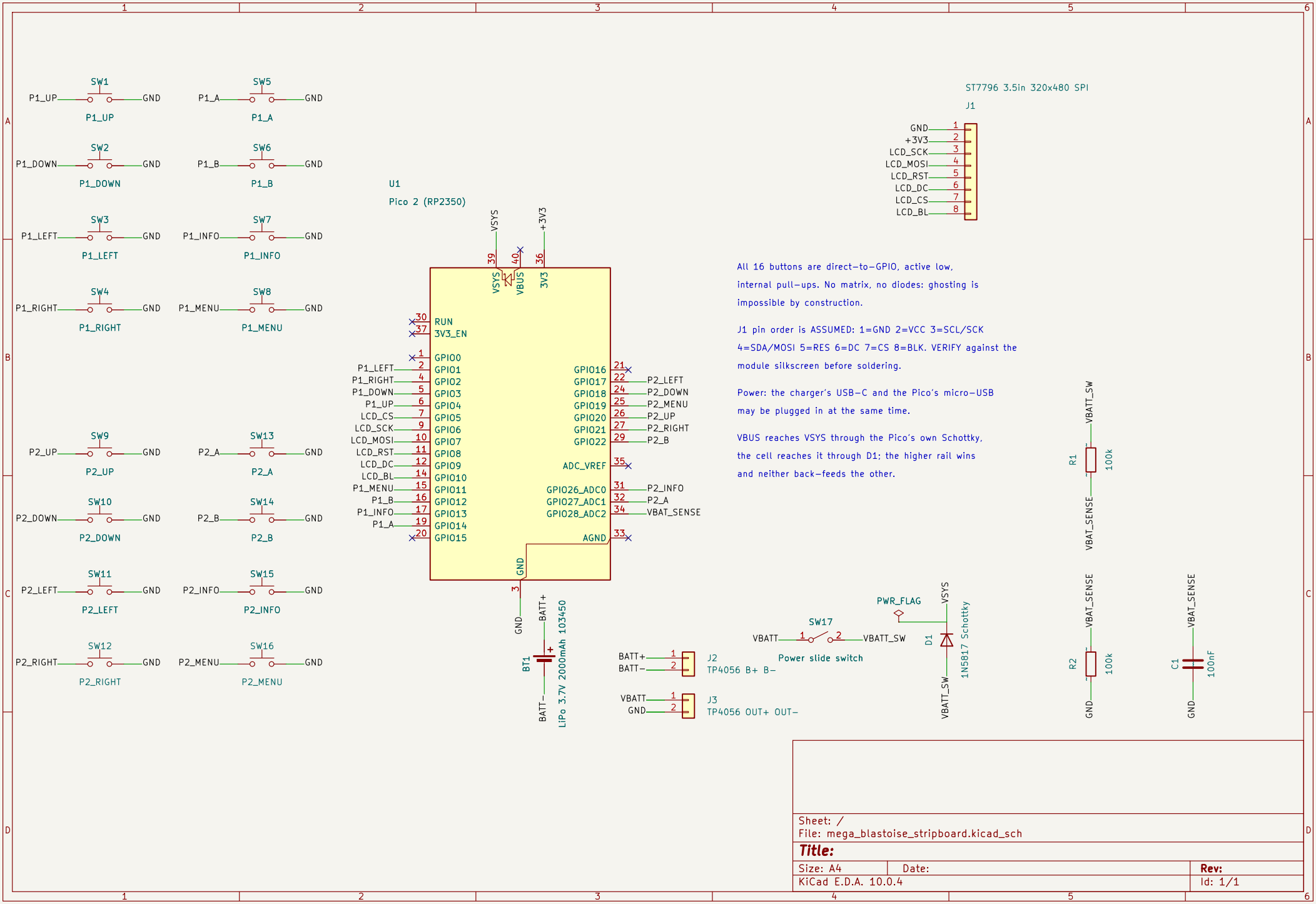

SW1
P1_UP
GND
P1_UP
SW5
P1_A
GND
P1_A
SW2
P1_DOWN
GND
P1_DOWN
SW6
P1_B
GND
P1_B
SW3
P1_LEFT
GND
P1_LEFT
SW7
P1_INFO
GND
P1_INFO
SW4
P1_RIGHT
GND
P1_RIGHT
SW8
P1_MENU
GND
P1_MENU
SW9
P2_UP
GND
P2_UP
SW13
P2_A
GND
P2_A
SW10
P2_DOWN
GND
P2_DOWN
SW14
P2_B
GND
P2_B
SW11
P2_LEFT
GND
P2_LEFT
SW15
P2_INFO
GND
P2_INFO
SW12
P2_RIGHT
GND
P2_RIGHT
SW16
P2_MENU
GND
P2_MENU
U1
Pico 2 (RP2350)
VSYS
+3V3
39
40
36
VSYS
VBUS
3V3
30
RUN
37
3V3_EN
1
GPIO0
P1_LEFT
2
GPIO1
P1_RIGHT
4
GPIO2
P1_DOWN
5
GPIO3
P1_UP
6
GPIO4
LCD_CS
7
GPIO5
LCD_SCK
9
GPIO6
LCD_MOSI
10
GPIO7
LCD_RST
11
GPIO8
LCD_DC
12
GPIO9
LCD_BL
14
GPIO10
P1_MENU
15
GPIO11
P1_B
16
GPIO12
P1_INFO
17
GPIO13
P1_A
19
GPIO14
20
GPIO15
21
GPIO16
P2_LEFT
22
GPIO17
P2_DOWN
24
GPIO18
P2_MENU
25
GPIO19
P2_UP
26
GPIO20
P2_RIGHT
27
GPIO21
P2_B
29
GPIO22
35
ADC_VREF
31
GPIO26_ADC0
P2_INFO
32
GPIO27_ADC1
P2_A
34
GPIO28_ADC2
VBAT_SENSE
33
AGND
GND
3
GND
BATT+
BT1
BATT−
LiPo 3.7V 2000mAh 103450
ST7796 3.5in 320x480 SPI
J1
GND
1
+3V3
2
LCD_SCK
3
LCD_MOSI
4
LCD_RST
5
LCD_DC
6
LCD_CS
7
LCD_BL
8
All 16 buttons are direct-to-GPIO, active low,
internal pull-ups. No matrix, no diodes: ghosting is
impossible by construction.
J1 pin order is ASSUMED: 1=GND 2=VCC 3=SCL/SCK
4=SDA/MOSI 5=RES 6=DC 7=CS 8=BLK. VERIFY against the
module silkscreen before soldering.
Power: the charger's USB-C and the Pico's micro-USB
may be plugged in at the same time.
VBUS reaches VSYS through the Pico's own Schottky,
the cell reaches it through D1; the higher rail wins
and neither back-feeds the other.
R1
VBATT_SW
100k
VBAT_SENSE
BATT+
1
BATT−
2
J2
TP4056 B+ B−
VBATT
1
GND
2
J3
TP4056 OUT+ OUT−
SW17
VBATT
1
2
VBATT_SW
Power slide switch
PWR_FLAG
D1
VSYS
1N5817 Schottky
VBATT_SW
R2
VBAT_SENSE
100k
GND
C1
VBAT_SENSE
100nF
GND
Sheet: /
File: mega_blastoise_stripboard.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 1/1